

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Moments

Moments questions become calm when you choose the pivot that deletes the worst unknown.

MOMENTS · M1 · 10

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The Map

This strategy booklet was mined from the local Mechanics 1 topic problem/answer PDFs, Dr Eslam's source notes, and the WME01 website answer bank. Mechanics questions become reliable when the diagram, sign convention, and equation family are chosen first.

01 PIVOT: Choose Pivot

TRIGGER *take moments about*

02 REACT: Find Reactions

TRIGGER *supports, reactions*

03 TILT: Tilting Limit

TRIGGER *on point of tilting, just lifts*

04 CENTRE: Centre Of Mass

TRIGGER *uniform rod, midpoint, centre*

BANK EVIDENCE Local pages: 39 problem, 57 answer, 13 notes. Website primary entries: 18. Website linked entries: 18.

01 PIVOT: Choose Pivot

TRIGGER *take moments about*

BECOMES Take moments about a point that removes an unknown reaction or tension.

FIRST LINE TO WRITE

$$\sum M_{\text{clockwise}} = \sum M_{\text{anticlockwise}}$$

SIMPLEST STRATEGY

1. Pick the pivot.
2. Ignore forces through pivot.
3. Value each perpendicular distance.
4. Orient clockwise/anticlockwise.
5. Tidy equation.

WORKED MODEL

▮ Taking moments about A removes the reaction at A.

HARD VARIANTS

1. Use perpendicular distance to the line of action.
2. Weight of a uniform rod acts at its midpoint.
3. For angled forces, use perpendicular component or perpendicular distance.

— BOTTOM LINE

A good pivot deletes one unknown.

COMMON TRAP Taking moments about a point that leaves every unknown.

02 REACT: Find Reactions

TRIGGER *supports, reactions*

BECOMES Use vertical equilibrium plus a moment equation.

FIRST LINE TO WRITE

$$R_A + R_B = \text{total weight}$$

SIMPLEST STRATEGY

1. Record upward reactions.
2. Equate vertical forces.
3. Apply one moment equation.
4. Calculate reactions.
5. Test by substitution.

WORKED MODEL

■ $R_A + R_B = 50g$, then moments gives one reaction.

HARD VARIANTS

1. Use mg for each mass.
2. If a support is at a pivot, its moment about that point is zero.
3. Check reactions are non-negative unless the support loses contact.

— BOTTOM LINE

Reactions are forces, so they belong in force balance and moments.

03 TILT: Tilting Limit

TRIGGER *on point of tilting, just lifts*

BECOMES At tilting, the reaction at the lifting support is zero.

FIRST LINE TO WRITE

$$R_{\text{lifting support}} = 0$$

SIMPLEST STRATEGY

1. Tell which pivot remains.
2. Ignore reaction at pivot in moments.
3. Let other support reaction be zero.
4. Take moments about pivot.

WORKED MODEL

■ On the point of tilting about C, the reaction at D is zero.

HARD VARIANTS

1. Greatest/least mass decides which end is about to lift.
2. Draw the pivot before writing equations.
3. Use equality at the limiting case, not inequality.

— BOTTOM LINE

Tilting means one support has just lost contact.

COMMON TRAP Taking moments about the support that has lifted.

04 CENTRE: Centre Of Mass

TRIGGER *uniform rod, midpoint, centre*

BECOMES Place the weight at the centre of mass before taking moments.

FIRST LINE TO WRITE

| uniform rod weight at midpoint

SIMPLEST STRATEGY

1. Check if body is uniform.
2. Establish midpoint/centre.
3. Name the weight.
4. Take distance from pivot.
5. Reuse in moments.
6. End with units.

WORKED MODEL

■ A uniform 6 m rod has weight acting 3 m from either end.

HARD VARIANTS

1. Non-uniform rods may give the centre distance as x .
2. If a particle is attached, its weight acts at its own point, not the rod midpoint.
3. Distances must be from the chosen pivot.

— BOTTOM LINE

Uniform means midpoint weight.

Practice Page: CENTRE

QUESTION TYPE Centre of mass

Place the weight of a uniform rod AB of length 5 m.

COMMON TRAP Putting the rod weight at a support.

Quick Recognition Table

WHEN YOU SEE	FIRST LINE TO WRITE
take moments about	$\sum M_{\text{clockwise}} = \sum M_{\text{anticlockwise}}$
supports, reactions	$R_A + R_B = \text{total weight}$
on point of tilting, just lifts	$R_{\text{lifting support}} = 0$
uniform rod, midpoint, centre	uniform rod weight at midpoint

Hard Variant Checklist

STRATEGY	CHECK BEFORE FINAL ANSWER
PIVOT: Choose Pivot	Use perpendicular distance to the line of action. Weight of a uniform rod acts at its midpoint. For angled forces, use perpendicular component or perpendicular distance.
REACT: Find Reactions	Use mg for each mass. If a support is at a pivot, its moment about that point is zero. Check reactions are non-negative unless the support loses contact.
TILT: Tilting Limit	Greatest/least mass decides which end is about to lift. Draw the pivot before writing equations. Use equality at the limiting case, not inequality.
CENTRE: Centre Of Mass	Non-uniform rods may give the centre distance as x . If a particle is attached, its weight acts at its own point, not the rod midpoint. Distances must be from the chosen pivot.